Parking Slot Reservation System

CMPE314 – Software Engineering Project Proposal

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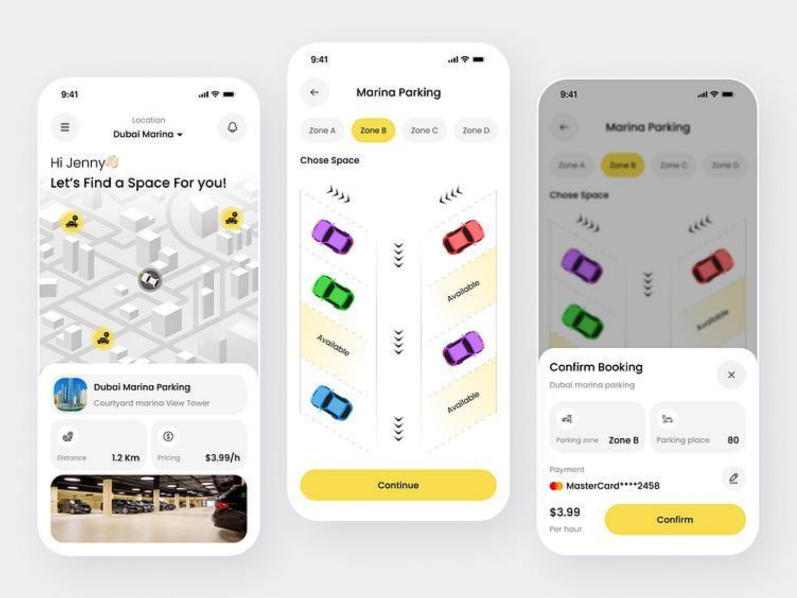


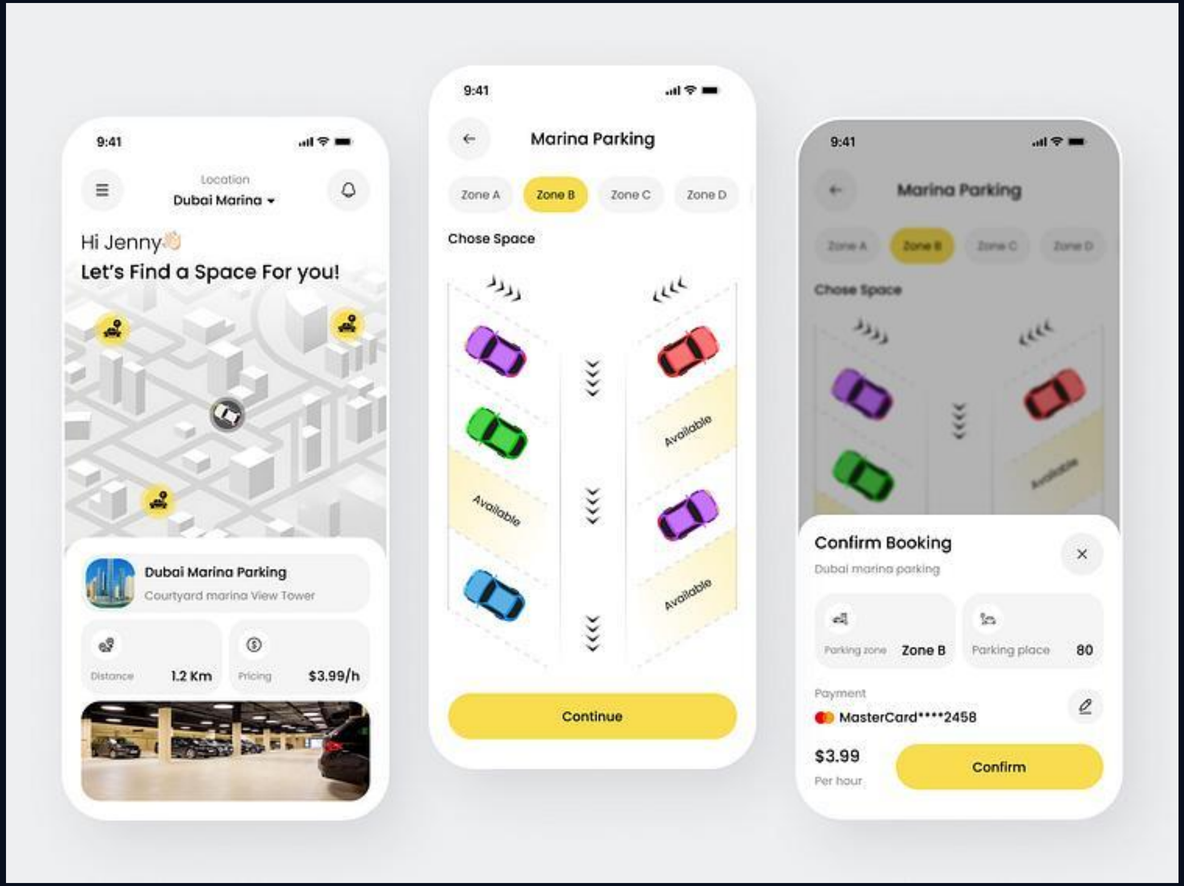
Presentation Outline

1. Introduction

Overview of the parking management challenge

Background

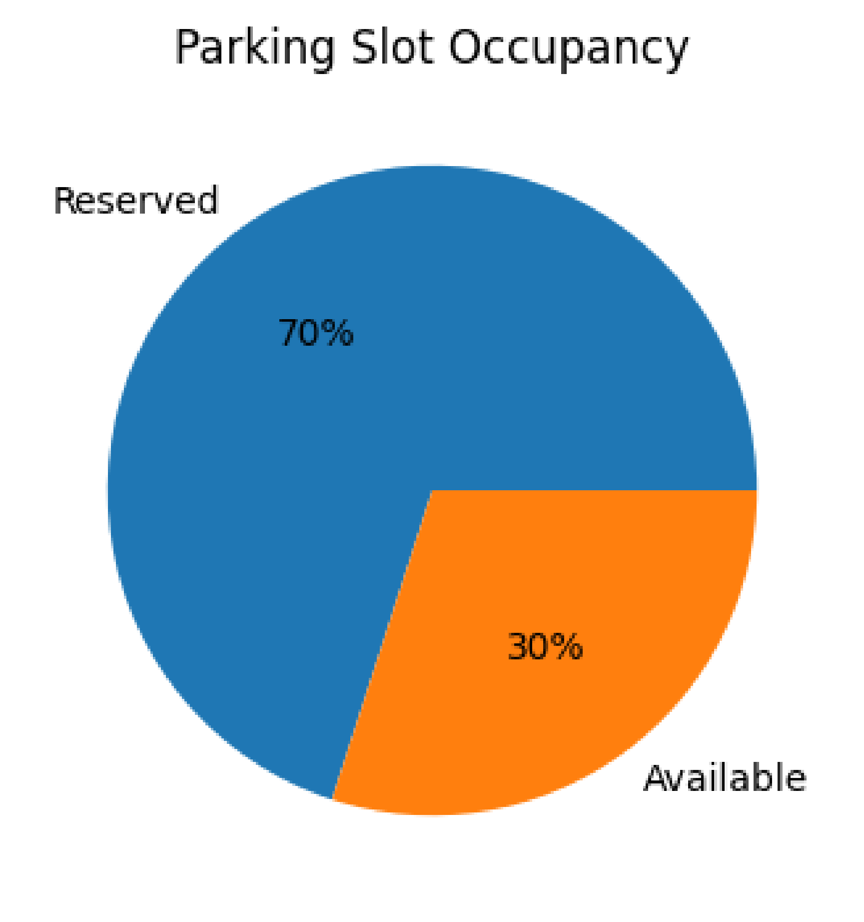
• Rapid increase in vehicle ownership has created parking challenges.
• Drivers spend unnecessary time searching for parking spaces.
• Manual parking systems lead to delays and congestion.



2. Problem Statement

Existing Issues

- No centralized parking reservation platform.
- Drivers cannot check slot availability remotely.
- Traffic congestion increases around parking areas.
- Inefficient slot management and poor reporting.



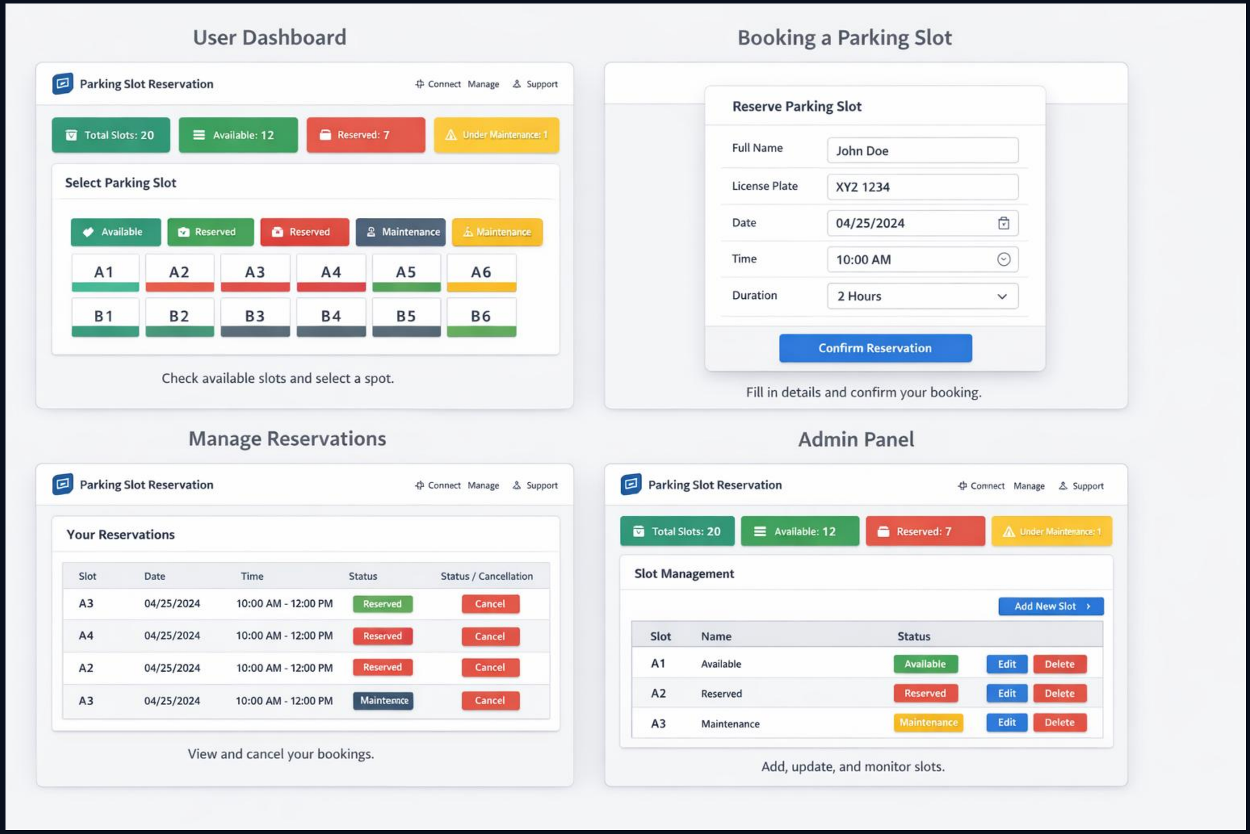
3. Project Aim & Objectives

Project Aim

- Develop a web-based system for reserving parking spaces efficiently.
- Improve user convenience and parking organization.

Objectives

- Provide secure user authentication.
- Enable online reservation and cancellation.
- Provide admin monitoring dashboard.
- Generate reports and notifications.



4. System Features

Authentication
- Secure registration and login system

Slot Availability
- Real-time parking slot updates

Reservation
- Reserve parking slots online and payment

Notifications
- Alerts and reminders

Admin Dashboard
- Manage slots and users

Reports
- Generate parking usage analytics

5. Functional Requirements

- FR-01: User Account Creation
- FR-02: Secure Login System
- FR-03: Display Available Parking Slots
- FR-04: Online Reservation System
- FR-05: Modify or Cancel Bookings
- FR-06: Notification Reminders
- FR-07: Admin Slot Management
- FR-08: Reports and Analytics

6. Technology Stack

Frontend
- HTML, CSS, JavaScript

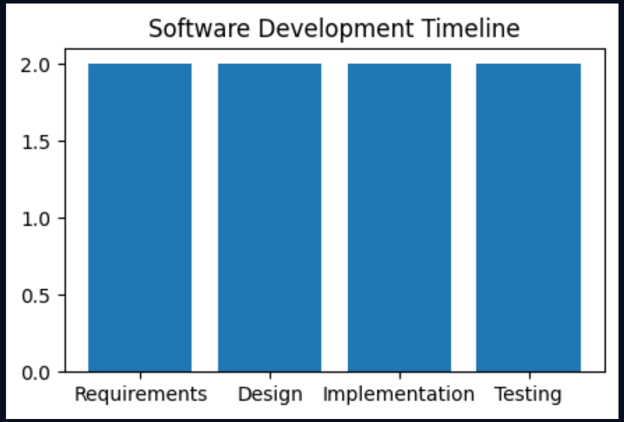
Backend
- Django / Node.js

Database
- MySQL / PostgreSQL

Version Control
- Git

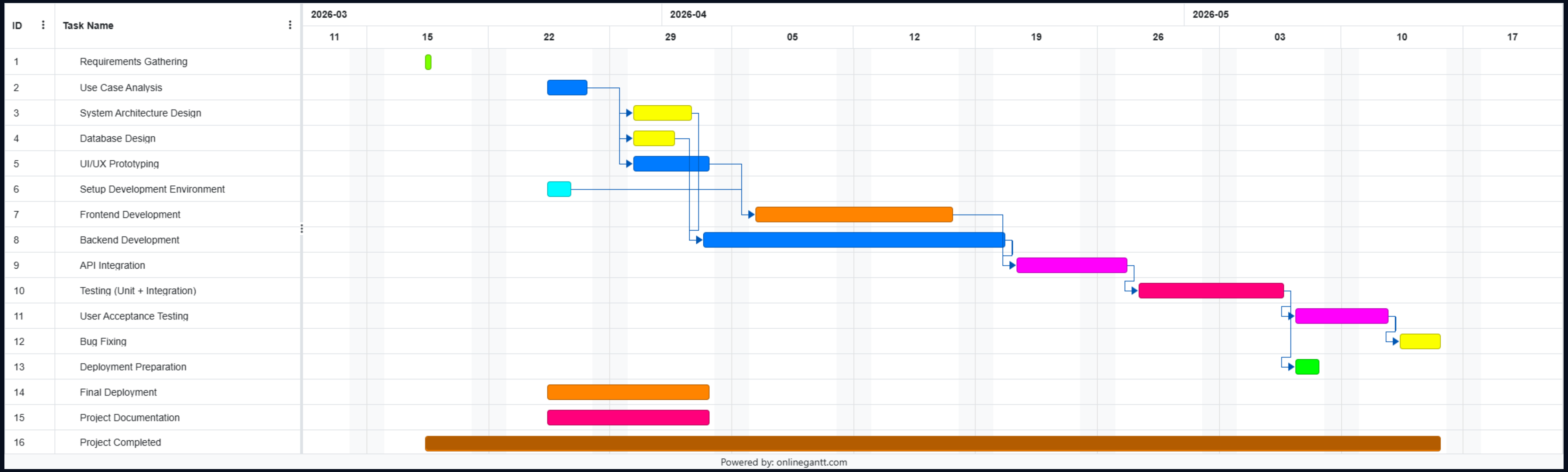
Libraries
- Bootstrap

7. Project Timeline and Gaant Chart



Development Phases

- Weeks 1–2: Requirements Gathering
- Weeks 3–4: UML Modeling & System Design
- Weeks 5–6: Core Feature Implementation
- Weeks 7–8: Testing, Documentation & Deployment



8. Expected Benefits

Benefits

- Reduces congestion and waiting time.
- Improves parking efficiency.
- Enhances user experience with online access.
- Provides data analytics for management decisions.
- Supports future scalability and smart parking integration.

9. Conclusion

Summary

- The Parking Slot Reservation System offers an efficient digital parking solution.
- The system demonstrates practical software engineering concepts.
- It improves organization, automation, and user convenience.
- Future improvements may include AI-based smart parking integration.